INCH-POUND
MIL-DTL-24135/13A(SH)
24 September 2020
SUPERSEDING
MIL-H-24135/13(SH)
3 September 1993

DETAIL SPECIFICATION SHEET

HOSE, SYNTHETIC RUBBER, HIGH PERFORMANCE, WIRE REINFORCED (DOUBLE-WIRE BRAID), FOR FLEXIBLE HOSE ASSEMBLIES (HIGH PRESSURE, HIGH TEMPERATURE, MULTI-BASED OIL, SIZES -4 THROUGH -32)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24135.

1. SCOPE

1.1 Scope. This specification covers high performance synthetic rubber hose, sizes -4 through -32, wire reinforced with a double-wire braid that are used in flexible hose assemblies for high pressure, high temperature, multi-based oil applications. This hose is compatible with petroleum, petroleum-based synthetic, and phosphate ester-based oils. This hose is not acceptable for use under vacuum.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-2104	-	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-H-19457	-	Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic
MIL-DTL-24135	-	Hose, Synthetic Rubber, Wire Reinforced, for Flexible Hose Assemblies, General Specifications for

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-DTL-24787 - Fittings, End, Reusable, for Flexible Hose Assemblies, General Specification for

MIL-DTL-24787/1 - Fittings, End, Types F, FL, and FFL (Flange), Reusable, for Flexible Hose Assemblies

MIL-DTL-24787/2 - Fittings, End, Type A (37-Degree Flare), Reusable, for Flexible Hose Assemblies

MIL-DTL-24787/3 - Fittings, End, Types C and CL (O-Ring Seal Union), Reusable, for Flexible Hose Assemblies

MIL-DTL-24787/5 - Fittings, End, Type D (Hose to Hose, 90-Degree Elbow), Reusable, for Flexible Hose Assemblies

MIL-DTL-32353 - Hydraulic and Lubricating Oil, Synthetic Hydrocarbon Base

MIL-PRF-83282 - Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Hose furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Material. Rubber compounds used for hose shall be resistant to petroleum- and phosphate ester-based oils and seawater. The compounds shall consist of synthetic rubber, or a blend of synthetic rubbers, and shall be in accordance with MIL-DTL-24135 and as specified herein.

3.3 Reinforcement. Wire reinforcement shall consist of a double-wire braid applied in such a manger as to meet all the requirements of this specification.

3.4 Dimensions and pressures. Hose dimensions shall be in accordance with table I and hose pressures shall be in accordance with table II.

TABLE I. Hose dimensions.

Hose size (dash number)	Hose dimensions (inches)						Minimum bend radius (inches)
	Inside diameter (ID) tube		Outside diameter (OD) reinforcement		OD cover		
	Min	Max	Min	Max	Min	Max	
-4	0.242	0.273	0.485	0.515	0.654	0.719	4
-6	0.367	0.398	0.649	0.681	0.812	0.875	5
-8	0.485	0.531	0.760	0.802	0.933	1.000	7
-10	0.610	0.656	0.889	0.921	1.058	1.125	8
-12	0.735	0.781	1.060	1.092	1.214	1.281	9.5
-16	0.985	1.040	1.353	1.399	1.511	1.609	12
-20	1.235	1.297	1.719	1.750	1.938	2.062	16.5
-24	1.485	1.547	1.969	2.047	2.188	2.312	20
-32	1.985	2.047	2.469	2.547	2.688	2.812	25

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TABLE II. Hose pressures and weight.

Hose size (dash number)	Pressure (psi)		Impulse peak pressure (psi)	Approximate weight/feet (lbs/ft)
	Working (max)	Burst (min)		
-4	5,000	20,000	5,800	0.31
-6	4,000	16,000	5,350	0.42
-8	3,500	14,000	5,000	0.51
-10	2,750	11,000	4,350	0.59
-12	2,250	9,000	3,950	0.75
-16	2,000	8,000	3,300	1.04
-20	1,625	6,500	2,950	1.68
-24	1,250	5,000	2,300	1.91
-32	1,125	4,500	2,000	2.39

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-DTL-24135 and 4.4.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-DTL-24135.

4.4 Sampling for qualification testing.

4.4.1 Hose. Hose samples for qualification testing shall be as specified in table III. Samples shall be not less than 15 inches in length and shall be fitted with end fittings qualified in accordance with MIL-DTL-24787.

TABLE III. Hose and end fitting samples required.

Hose size (dash number)	Number of hose samples to be provided	Number of end fitting samples to be provided 1/
-8	6	12
-32	7 2/	14

NOTES:

1/ End fittings for testing shall be qualified in accordance with MIL-DTL-24787.

2/ Ensure there is an adequate amount of hose to conduct flame spread testing (see 4.5.11 of MIL-DTL-24135).

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4.4.2 End fittings. End fittings used for the qualification of hose to this specification sheet shall be end fittings qualified in accordance with MIL-DTL-24787, using any of the end fitting types identified in table IV. The quantity of end fittings required shall be in accordance with table III. The end fitting supplier and hose supplier may be a single source. If a combined end fitting and hose qualification is pursued, end fittings in accordance with MIL-DTL-24787/2 or /3 shall be used for the hose qualification portion to assist with end fitting qualification. Upon receiving qualification of this hose, the hose shall be considered qualified for use with all end fitting types identified in table IV.

TABLE IV. MIL-DTL-24787 end fittings for use with MIL-DTL-24135/13 hoses.

End fitting specification sheet number	End fitting type per specification sheet	End fitting description
24787/1 1/	F, FL, FFL	Flange to hose (straight, 90-degree elbow, or 45-degree elbow)
24787/2	A	37-degree flare swivel
24787/3 2/	C, CL	O-ring seal union (straight or 90-degree elbow)
24787/5 2/	D	90-degree elbow

NOTES:
1/ For -12 to -32 hose dash sizes only.
2/ Does not include -10 hose dash sizes.

4.4.2.1 Hose and end fitting mix. Where there are two or more qualified suppliers of end fittings in accordance with MIL-DTL-24787, or when the prospective hose supplier is also qualifying an end fitting, the hose and end fitting mix shall be in accordance with 4.4.2.1.1 through 4.4.2.1.3 below, as appropriate. When qualifying a hose and end fitting, hose samples shall be coupled on one end with the end fitting being submitted for qualification and on the other end with a previously qualified end fitting.

4.4.2.1.1 Qualifying hose with multiple qualified end fitting suppliers. Where there are two or more qualified suppliers of end fittings, the end fitting mix shall be in accordance with table V.

TABLE V. End fitting mix when there are multiple qualified end fitting suppliers for hose qualification test.

Sample size 1/ (dash number)	Number of end fittings to be provided 2/	
	Qualified source #1	Qualified source #2
-8	8	4
-32	10	4

NOTES:
1/ Satisfactory completion of qualification testing of the samples selected in accordance with tables I, II, and III will qualify the manufacturer for all sizes in this specification sheet for the end fitting specification sheet(s) in table IV.
2/ See table IV for end fitting options.

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4.4.2.1.2 Qualifying end fitting and hose with one qualified end fitting supplier and qualified hose supplier. Where the prospective hose supplier is also qualifying an end fitting, and there is only one qualified supplier of end fittings, hose, or both, the end fitting and hose mix shall be in accordance with the section of the same title in MIL-DTL-24787/2 or /3. For end fittings being qualified at the same time, the sequence of tests for end fitting qualification shall be in accordance with MIL-DTL-24787/2 or /3. The sequence of tests for hose qualification shall be in accordance with 4.5. The sample size of hose dash of -8 represents the smallest sample size in accordance with MIL-DTL-24787/2 or /3, while hose dash of -32 represents the largest sample size.

4.4.2.1.3 Qualifying end fitting and hose with two or more qualified end fitting and qualified hose suppliers. Where the prospective hose supplier is also qualifying an end fitting, and there are two or more qualified suppliers of end fittings, hose, or both, the end fitting and hose mix shall be in accordance with the section of the same title in MIL-DTL-24787/2 or /3. For end fittings being qualified at the same time, the sequence of tests for end fitting qualification shall be in accordance with MIL-DTL-24787/2 or /3. The sequence of tests for hose qualification shall be in accordance with 4.5. The sample size of hose dash of -8 represents the smallest sample size in accordance with MIL-DTL-24787/2 or /3, while hose dash of -32 represents the largest sample size.

4.5 Test sequence. The sequence of tests on sample hose assemblies shall be as specified in table VI.

TABLE VI. Sequence of qualification tests on sample assemblies.

Test sequence	Sample numbers												
	Size -8						Size -32						
	1	2	3	4	5	6	7	8	9	10	11	12	13
Proof pressure (see 4.5.1 1/)	X	X	X	X	X	--	X	X	X	X	X	--	--
Stability (see 4.5.2 1/)	X	X	X	X	X	--	X	X	X	X	X	--	--
Aging (see 4.5.3 1/)	X	X	X	--	--	--	X	X	X	--	--	--	--
Oil circulation (see 4.7 2/)	--	X	--	--	--	--	X	--	--	--	--	--	--
Impulse (see 4.5.5 1/)	--	--	--	X	--	--	--	--	X	--	--	--	--
Burst (see 4.5.6 1/)	X	--	--	--	--	--	--	--	--	--	X	--	--
Cold temperature flexibility (see 4.5.8 1/)	--	--	X	--	--	--	--	X	--	--	--	--	--
Fire resistance (see 4.5.10 1/)	--	--	--	--	--	X	--	--	--	--	--	X	X

NOTES:
1/ See specified section in MIL-DTL-24135 for test details.
2/ See specified section within this specification sheet.

4.6 Test fluid. The test fluid shall be oil or hydraulic fluid in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-DTL-32353, MIL-PRF-83282, or MIL-H-19457 for the aging, oil circulation, impulse, and cold temperature tests.

4.7 Oil circulation. Sample hose assemblies shall be filled with hydraulic fluid in accordance with4.6. Assemblies shall be installed in a 180-degree arc and tested on a flex and circulation test stand as shown on figure 1. The radius at the inside of the bend shall be equal to the applicable arc and flex bend radius specified in table I. Oil maintained at +300±5 °F shall be circulated through the assemblies at 60 psi, at aflow rate for note less than 3 gallons per minute, while the assemblies are flexed 3.5 inches in each direction at 12±1 cycles per minute (CPM). The hose assembly shall be subjected to 400 hours of hot oil circulation and flexing without evidence of leakage or other malfunction. The oil shall be changed after 200 hours of circulation.

4.8 Impulse pressure. The peak pressure for the dynamic impulse test shall be to the impulse peak pressure shown in table II. Impulse test for 300,000 cycles at 300 °F using the pressure time curve on figure 2.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hoses are intended for use in piping systems as flexible connections to resiliently mounted (sound mounted) equipment that may be used for other applications, which do not exceed the pressure or temperature limits of the hose selected, and have an established service life as documented in S6430-AE-TED-010, Piping Devices, Flexible Hose Assemblies. Hose and hose assemblies procured to this specification are form, fit, and function replacements for prior revisions. Existing stock of hose and hose assemblies manufactured to MIL-H-24135B, dated 3 September 1993, are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Size of hose (see 3.4).

c. Quantity of hose (in linear feet) per size.

d. Type and size of end fitting for hose assembly (see table IV and MIL-DTL-24787).

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24135 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

37-degree flare swivel to hose, straight

90-degree elbow, hose to hose

Flange to hose, 45-degree elbow

Flange to hose, 90-degree elbow

Flange to hose, straight

O-ring seal union, male half to hose, 90-degree elbow

O-ring seal union, male half to hose, straight

Petroleum

Phosphate ester

Seawater

Synthetic oil

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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HOSE CUT LENGTH (HCL) = R+2J+3.5+20D

$R = r + \frac{OD}{2}$

r = MINIMUM BEND RADIUS

J = SOCKETED HOSE

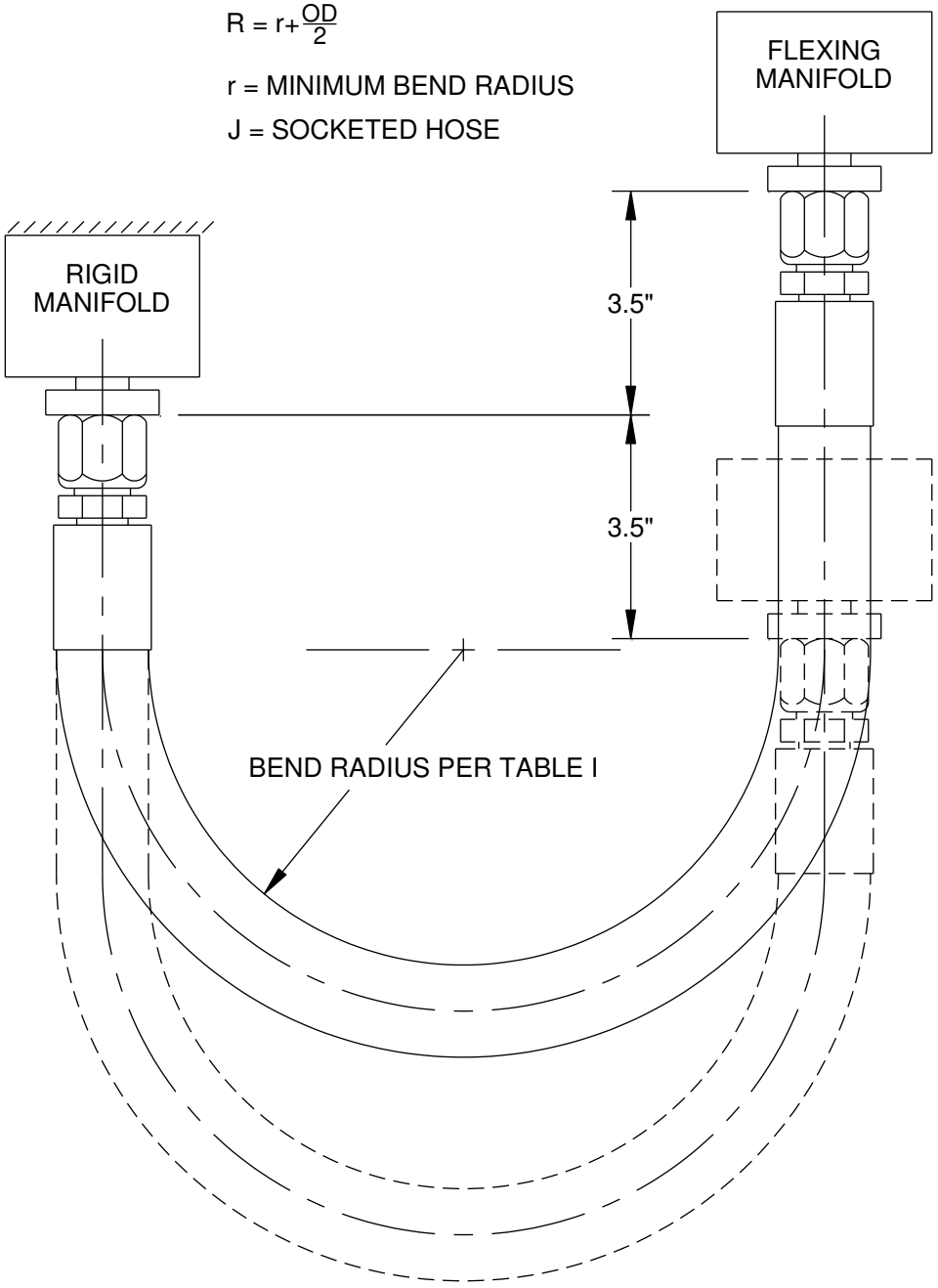
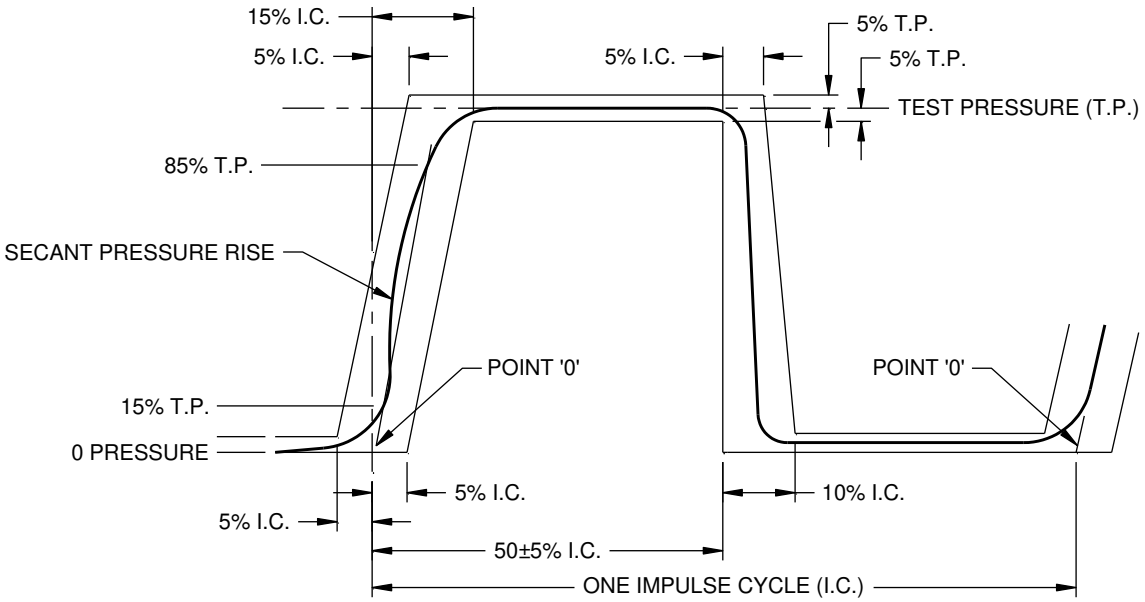


FIGURE 1. Flex and circulation test stand.

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NOTES:

1. SECANT PRESSURE RISE – The straight line drawn through two points on the pressure rise curve, one point at 15% of the test pressure and the other at 85% of the test pressure.
2. SECANT PRESSURE RISE shall be targeted at 11.7 × Test Pressure with a tolerance of ±10 percent.
3. Point “0” – The intersection of the SECANT PRESSURE RISE with a 0 pressure.

FIGURE 2. Impulse pressure curve.

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CONCLUDING MATERIAL

Preparing activity:
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